U4RASD at StanceEval-2026: Fine-Tuning, Prompting, and Ensembling for Unseen-Target Arabic Stance Detection

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Abstract

This paper describes our system for StanceEval 2026 Track 2, which requires predicting stance toward targets not seen during training. We explored fine-tuning different Arabic BERT encoder models, using class weighting techniques, incorporating auxiliary sentiment and sarcasm tasks in sequential and parallel multi-task setups, and focal and margin-based contrastive losses. We also tried zero-shot and few-shot prompting, self-consistency with escalation, LoRA fine-tuning, and an ensemble approach. Our best-performing submission on the official blind test set was a training-free few-shot prompting approach using Gemini 2.5 Flash with an extended prompt, achieving an F_{avg2} of 87.41 and an F_{avg3} of 76.94, and ranking 8th out of 13 teams. This result shows that a simpler, training-free prompting approach can generalize to unseen targets more effectively than fine-tuned and ensemble systems.

1 Introduction

Stance detection is the task of determining whether the author of a piece of text is in favor of, against, or has no stated opinion toward a specified target (Somasundaran and Wiebe, 2010; Mohammad et al., 2016). Unlike sentiment analysis, which assigns a polarity to a text on its own, stance detection requires reasoning about the relationship between the text and a specific target: the same piece of text can express different stances toward different targets, and the target itself is not necessarily mentioned explicitly in the text (Mohammad et al., 2016). This makes the task sensitive to implicit or sarcastic opinions and distinguishes it from standard sentiment classification. Beyond its use as an NLP benchmark, stance detection supports the analysis of public opinion and polarization on social media, complementing traditional survey-based methods (Küçük and Can, 2020).

This paper describes our submission to Track 2 at StanceEval 2026 (Albalawi et al., 2026a), which focuses on classifying the stance of Arabic social media text toward a target that is not seen during training, requiring models to generalize their notion of stance rather than memorize target-specific features. We investigate this problem along four directions: fine-tuning encoder-based Arabic language models, incorporating auxiliary objectives through custom loss functions and multi-task learning, prompting and fine-tuning large language models (LLMs), and combining the strongest individual models into an ensemble. Our strongest system on the development set was a majority-vote ensemble of three individual models (F_{avg2} 89.59), but it fell to 79.39 on the blind test, while our few-shot prompting system using an extended prompt scored 88.74 on development and 87.41 on the blind test set, ranking 8th out of 13 teams.

2 Related Work

Arabic stance detection datasets and shared tasks. Arabic stance detection was first formalized as a shared task on the Mawqif dataset (Alturayeif et al., 2022, 2024). A later shared task extended it to a cross-topic, political-domain setting (Aldous et al., 2026).

Auxiliary objectives and multi-task learning. Multi-task learning has been used to improve stance detection in several ways. One line of work trains a single model jointly across multiple target pairs, rather than one model per target, to learn more generalizable representations and reduce overfitting (Li and Caragea, 2021). Closer to our work, auxiliary sentiment and sarcasm objectives have been used for Arabic stance detection (Almasoudi et al., 2025), motivating our sequential architecture (Section 4.2.2). The top system in the previous StanceEval shared task combined a multi-loss contrastive ensemble with multi-task learning (Badran et al., 2024), but evaluated on seen targets, unlike our setting.

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3 Data

We use the Track 2 partition of the Mawqif-XT dataset (Alturayeif et al., 2022; Albalawi et al., 2026b). In addition to the target, text, and gold stance label, each sample includes auxiliary sarcasm and sentiment annotations, which we use in our multi-task learning experiments (Section 4.2.2). Table 2 (Appendix A) reports the label distribution for the Track 2 training and development sets.

4 System

We approach Track 2 along four directions: fine-tuning encoder-based Arabic language models as a baseline, incorporating auxiliary training objectives on top of the strongest baseline, prompting and fine-tuning LLMs, and combining our best individual systems into an ensemble.

4.1 Baseline

We fine-tuned several BERT-based Arabic encoders without any auxiliary objectives: AraBERTv0.2-Twitter-large (Antoun et al., 2020), MARBERT (Abdul-Mageed et al., 2021), CAMeLBERT-MSA (Inoue et al., 2021), and ModernAraBERT (Giza Systems Data Science Team, 2025). AraBERTv0.2-Twitter-large achieved the strongest baseline performance.

4.2 Auxiliary Training Objectives

Starting from our strongest baseline, AraBERTv0.2-Twitter-large, we explored two families of auxiliary training objectives: alternative loss functions and weighting techniques, and multi-task learning with auxiliary sentiment and sarcasm heads.

4.2.1 Loss Functions, Weighting Techniques, and Learning Rate Scheduling

We experimented with alternative loss functions, replacing the default cross-entropy loss with focal loss (Lin et al., 2017) and a margin-based contrastive classification loss (Hadsell et al., 2006). We also tested class-weighting techniques, including class balancing (Cui et al., 2019) and inverse class frequency. Our best configuration in this family paired AraBERTv0.2-Twitter-large with the contrastive classification loss (margin $= 0.5$, $\lambda = 1.0$).

4.2.2 Multi-Task Learning

We experimented with two multi-task learning approaches that integrate auxiliary sentiment and sarcasm objectives alongside stance classification: a sequential architecture (Almasoudi et al., 2025) and a parallel architecture.

In the **sequential** architecture, each auxiliary head is applied before the stance head, and every subsequent head receives the shared [CLS] embedding concatenated with the logits of all preceding tasks. Our best sequential configuration integrated a sentiment head before the stance head, excluding a sarcasm head, and used a per-task contrastive loss (margin $= 0.5$, $\lambda = 1.0$) for both heads.

In the **parallel** architecture, the sarcasm, sentiment, and stance heads are applied independently on the shared [CLS] embedding, and their outputs, together with the raw stance logits, are combined by a unified fusion classifier to produce the final stance prediction. Our best parallel configuration weighted the task losses as 0.1 (sarcasm), 0.2 (sentiment), and 0.7 (stance), applying class-balanced weighting ($\beta = 0.99$) to the sarcasm and stance losses (Cui et al., 2019).

The sequential and parallel architectures are illustrated in Figures 1a and 1b, respectively (Appendix E).

4.3 Prompting Approaches

We experimented with different prompting-based approaches that varied in the prompt and the inference strategy used. We constructed a baseline prompt, listed in Appendix C which was designed based on the official annotation guidelines released with the Mawqif dataset and which we used with no in-context examples. We also constructed an extended prompt, listed in Appendix D, which keeps the same label definitions from the baseline prompt but extends the prompt with additional decision rules, eight in-context examples, and requests a structured output format from the model.

4.3.1 Zero-Shot Prompting

We evaluated zero-shot prompting using GPT-4o-mini (OpenAI, 2024), with the baseline prompt in Appendix C and no in-context examples.

4.3.2 Zero-Shot with Self-Consistency and Escalation

We experimented with zero-shot prompting that combines self-consistency (Wang et al., 2023) with an escalation step to a stronger model under uncertainty. For each sample, a student model, GPT-4o-mini (OpenAI, 2024), generated five independent predictions at temperature 0.7, using the baseline prompt. Each generation was required to produce

ID	System	F_{avg2}	F_{avg3}
5	Contrastive loss (AraBERTv0.2-Twitter-large)	75.69	58.65
10	**Few-shot (Gemini 2.5 Flash)**	**87.41**	**76.94**
14	LoRA: Qwen2.5-14B	76.58	68.12
15	Ensemble (6+9+14)	79.39	70.42

Table 1: Blind test set performance for our four submitted systems. IDs match Table 3. Best result in bold.

chain-of-thought reasoning (Wei et al., 2022) before outputting a label; we found this more effective than greedy decoding. The five labels were aggregated by majority vote, and samples with less than 80% agreement (fewer than four of five predictions) were escalated to a teacher model, Claude Sonnet 4.6 (Anthropic, 2026), which generated three predictions at temperature 0.3; their majority label was used as the final prediction.

4.3.3 Few-Shot Prompting with an Extended Prompt

We evaluated few-shot prompting Gemini 2.5 Flash (Comanici et al., 2025) using the extended prompt. The prompt states the three label definitions from the baseline prompt and includes additional notes to guide the model's interpretation and covers cases we observed in the training data. One note states that the stance is judged toward the target itself rather than toward events concerning it, so that supporting a target's cancellation counts as *Against* and opposing it counts as *Favor*. A second rule separates stance from sentiment and tone, where a tweet may be negative in tone yet *Favor* its target, and may be positive or joking yet be *Against* it. The rest of the notes alert the model for the possibility of implicitly mentioned targets, differing dialects, and humor. The prompt includes eight labeled examples randomly drawn from the training split. The model is instructed to return the label with a short rationale and a confidence score.

4.4 Fine-tuning

We explored LoRA fine-tuning (Hu et al., 2022) on several instruction-tuned LLMs: Qwen2.5-14B-Instruct (Yang et al., 2024), Gemma3-12B (Gemma Team et al., 2025), Command R7B Arabic (Alnumay et al., 2025), and Llama-3.1-8B-Instruct (Llama Team, 2024).

4.5 Ensemble

We developed an ensemble that combines the strongest system from each of the three methodologies we experimented with, as measured on the development set: the sequential multi-task model (Section 4.2.2), the LoRA fine-tuned Qwen2.5-14B-Instruct model (Section 4.4), and the self-consistency-with-escalation approach (Section 4.3.2). The few-shot system (Section 4.3.3) was developed after the ensemble had been submitted and so did not enter this selection.

Before combining the three models, we examined the correlation of prediction errors across them to assess how many mistakes were shared versus complementary, measuring correlation using each model's per-instance correctness – that is, whether its prediction matched the gold label. Figure 2a and Figure 2b (Appendix F) report the pairwise correlation of correctness and the proportion of rows on which each pair of models failed simultaneously, respectively. Guided by this analysis, we adopted majority voting as our final ensembling strategy, with ties resolved in favor of the *Favor* label.

5 Experimental Setup

Data. We train on the official Track 2 training split (2,721 samples) and validate on the development split (1,400 samples), whose target is women's empowerment; the blind test set covers two further unseen targets, Ecars and Trimester.

Preprocessing. Encoder models' inputs are stripped of diacritics, tatweel, and all characters outside the Arabic block except digits, with runs of three or more identical characters collapsed to two. The prompting-based methods and LoRA fine-tuning use the raw text.

Metrics. We use F_{avg2}, the official task ranking metric, as our model selection criterion. It macro-averages the F_1 of *Favor* and *Against*; F_{avg3} averages over all three classes.

Encoder training. Encoders use the hyperparameters in Tables 5 and 6, trained for up to 12 epochs with early stopping on development F_{avg2} (patience 5). Our contrastive-loss configuration (Section 4.2.1) additionally uses a linear scheduler (warmup ratio 0.1), batch size 16, and a learning rate of 1×10^{-5}.

LLM inference. Prompting approaches use default decoding parameters except temperature: 0 for zero-shot, 0.7 for self-consistency, and 0.3 for teacher escalation and few-shot prompting. We

accessed all models through the OpenRouter[1] platform.

LoRA fine-tuning. Models were trained on data pairing each gold label with a reasoning trace from Claude Sonnet 4.6 (Anthropic, 2026), using the configuration in Table 7, selected by varying one hyperparameter at a time.

Implementation. All experiments ran on a single NVIDIA A100 GPU. All models were trained using HuggingFace Transformers (Wolf et al., 2020), using a fixed seed of 42.

6 Results and Analysis

Table 3 (Appendix B) reports development results for all configurations. The ensemble (ID 15) achieved the best development F_{avg2} score of 89.59. Table 1 reports results for our four blind-test submissions.

Blind test results. The fine-tuned AraBERTv0.2-Twitter-large model with contrastive loss (ID 5), Qwen2.5-14B-Instruct with LoRA (ID 14), and our ensemble (ID 15) achieved F_{avg2} scores of 75.69, 76.58, and 79.39, respectively, compared with 82.86, 88.88, and 89.59 on the development set. In contrast, few-shot prompting with the extended prompt (ID 10) achieved 87.41, only 1.33 points below its development score of 88.74. This contrasts with the drops of 7.17 to 12.30 points for the systems involving training. Despite ranking third on the development set, the few-shot system outperformed our other submissions on the blind test by 8.0 to 11.7 F_{avg2} points, showing substantially better generalization.

Error Analysis Our two best-performing systems, the few-shot extended-prompt system and the ensemble, were substantially more accurate when sentiment matched the intuitive direction of the gold stance (Favor+Positive, Against+Negative, or None+Neutral). The ensemble misclassified 6.0% of matched rows versus 30.1% of mismatched rows, and the few-shot system showed a similar pattern (6.1% vs. 33.2%). Sarcastic tweets, a subset of this mismatched category, were likewise more difficult (28.6% vs. 11.3% errors for the ensemble and 30.4% vs. 12.1% for few-shot prompting). Table 4 provides the full breakdown. The consistency of this pattern across both systems suggests that these cases remain challenging regardless of the underlying modeling approach.

7 Conclusion

In this paper, we presented our system for Track 2 of StanceEval 2026, which focuses on stance prediction toward unseen targets. We explored fine-tuning Arabic BERT models, class weighting, auxiliary sentiment and sarcasm tasks, contrastive losses, prompting, LoRA fine-tuning, and ensembling. Our best-performing submission was a training-free few-shot prompting approach using Gemini 2.5 Flash, achieving an F_{avg2} of 87.41 and an F_{avg3} of 76.94 on the official blind test set. This result suggests that training-free prompting can generalize to unseen targets more effectively than the fine-tuned and ensemble systems we evaluated.

Limitations

Our work has the following limitations. First, ties in our final ensemble were resolved in favor of the *Favor* label, since it is the majority class in the development set (894 of 1,400 samples, or 63.9%); this choice may still bias ambiguous predictions toward *Favor* on data with a different class distribution.

Second, our self-consistency-with-escalation approach, while offering a more stable F_{avg2} than a single greedy LLM call, comes at a considerably higher inference cost, requiring up to five student calls and three teacher calls per sample.

Finally, the Track 2 development split contains a single target, women's empowerment, so all of our model selection rested on one unseen target. The development-to-test gaps we report are consistent with this being a limitation of the evaluation setup rather than of any individual system.

Ethics Statement

Our work uses the Mawqif-XT dataset (Albalawi et al., 2026b), which was released publicly by the shared task organizers under their terms of use; we collected no new data and performed no additional annotation, and we did not attempt to identify the authors of any tweet. Our prompting-based methods rely on proprietary models whose training data is undisclosed, so we cannot fully characterize the biases they may carry into stance predictions and any downstream deployment should account for this.

[1] https://openrouter.ai/

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Topic	Split	Favor	Against	None
COVID-19 Vaccine	Train	599	598	176
Digital Transformation	Train	1,035	167	146
Women's Empowerment	Dev	894	436	70
Total		2,528	1,201	392

Table 2: Track 2 label distribution by topic. Train and Dev are disjoint by topic (unseen-target setting).

ID	System	F_{avg2}
1	ModernAraBERT	67.24
2	CAMeLBERT-MSA	77.66
3	MARBERT	79.73
4	AraBERTv0.2-Twitter-large	82.40
5	Contrastive loss (AraBERTv0.2-Twitter-large) (§4.2.1)	82.86
6	Sequential multi-task (AraBERTv0.2-Twitter-base) (§4.2.2)	83.14
7	Parallel multi-task (AraBERTv0.2-Twitter-base) (§4.2.2)	81.56
8	Zero-shot (GPT-4o-mini) (§4.3.1)	84.80
9	Zero-shot with Self-consistency and escalation (Claude Sonnet 4.6 + GPT-4o-mini) (§4.3.2)	88.15
10	Few-shot (Gemini 2.5 Flash) (§4.3.3)	88.74
11	LoRA: Llama-3.1-8B-Instruct	83.64
12	LoRA: Command R7B Arabic	85.32
13	LoRA: Gemma3-12B	88.74
14	LoRA: Qwen2.5-14B-Instruct	88.88
15	**Ensemble (6+9+14)**	**89.59**

Table 3: Development-set F_{avg2} for every system evaluated. Blue rows were submitted to the blind test evaluation (Table 1). Best result in bold.

A Track 2 Label Distribution

B Development Set Results

C Baseline Prompt Template

```
You are an expert Arabic linguist specializing in Stance Detection.
Your task is to determine the author's stance toward a specific target based on a tweet.
### Annotation Guidelines:
You must classify the stance as one of the following three labels: [Favor, Against, None].
1. Favor:
   - The author explicitly supports the target or something aligned with it.
   - OR the tweet contains information (news, quotes) that reveals the author supports
     the target, even if not explicitly stated.
2. Against:
   - The author is explicitly against the target or something aligned with it.
   - OR the tweet contains information (news, quotes) that reveals the author is against
     the target, even if not explicitly stated.
3. None:
   - There is no evidence in the tweet to judge the stance (e.g., neutral news, simple
     inquiries).
   - OR the tweet is irrelevant to the target.
### Important Rules:
- Evaluate the author's intent.
- Do not use your own personal opinion.
- If the tweet is sarcastic, look for the implicit meaning (e.g., positive words used
  to criticize = Against).
- Output ONLY the label: Favor, Against, or None.
```

D Extended Prompt Template

SYSTEM PROMPT:
You are an expert analyst of Arabic social media discourse specializing in stance detection: identifying the author's position toward a target, which is distinct from the sentiment or tone of the text. You are fluent in Modern Standard Arabic (MSA) and various regional dialects (e.g., Khaleeji, Levantine, Egyptian, Maghrebi).

USER PROMPT TEMPLATE:
Determine the author's stance toward the target `"{topic}"` in the provided Arabic tweet. Imagine you were the author of the tweet and ask: is my intention to support or be against the target?

Labeling Rules (from the official annotation guidelines)

1. **favor** — it can be inferred that the author supports the target, either because:
   - The tweet explicitly supports the target or something aligned with the target, OR
   - The tweet is not overtly supporting/against anything, but contains information (news, quote, story, etc.) that reveals the author supports the target.
2. **against** — it can be inferred that the author is against the target, either because:
   - The tweet explicitly opposes the target or something aligned with the target, OR
   - The tweet is not overtly supporting/against anything, but contains information (news, quote, story, etc.) that reveals the author is against the target.
3. **none** — the tweet provides no hint as to the author's stance, either because:
   - There is no evidence to judge the stance (e.g., inquiries, news that expresses no positive or negative position), OR
   - The tweet is irrelevant to the target.

Additional Notes

- **Anchor the stance to the target itself, not to events about it.** Supporting the cancellation, removal, reversal, or replacement of the target means the stance is AGAINST the target (e.g., celebrating or demanding that the target be abolished = against). Opposing its cancellation means FAVOR. Ask: does the author want the target to exist/continue, or not?
- **Stance is NOT sentiment.** Judge the author's position toward the target, not the tone of the tweet. A tweet can be angry or negative in tone yet FAVOR the target (e.g., complaining that the target is delayed, obstructed, or not implemented enough), and a tweet can sound positive or joking yet be AGAINST the target (e.g., sarcasm, mock praise).
- Judge the author's intent only; ignore your own opinion of the topic.
- The target may be mentioned explicitly or only implicitly.
- Watch for sarcasm and irony: Arabic social media often uses "black humor" where the literal polarity is the opposite of the intended stance.
- Interpret regional dialects (Khaleeji, Levantine, Egyptian, Maghrebi) correctly.

Labeled Examples
The following examples from other targets illustrate the labeling convention:

- Target: Digital Transformation | Text: "26 مليون كتاب !! اين التحول الالكتروني للمناهج ؟ كمية هدر سنوي للكتب مؤسفة نتمنى احلال الاجهزة اللوحية بدلاً من الكتب" → favor (tone is angry/negative, but the author is DEMANDING digital transformation — negative sentiment, favor stance)
- Target: Digital Transformation | Text: "وزارة الاتصالات تأخرت في التحول الالكتروني العالم الان يعمل على الخدمات الذكية والرقمية" → favor (criticizes the ministry for being LATE to digitalize — the complaint implies support for the target itself)
- Target: Covid Vaccine | Text: "شلل بسيط بسبب تطعيم كورونا عادي ليش مكبرين الموضوع" → against (joking/light tone, but the sarcastic mock-minimization of vaccine side effects attacks the vaccine — against)
- Target: Covid Vaccine | Text: "هههه لسا ما اقتنعت انه في كورونا عشان اخذ تطعيم" → against (laughing tone, but the author rejects taking the vaccine — against)
- Target: Digital Transformation | Text: "التحول الالكتروني وفر وقت وجهد وكل شيء" → favor (straightforward praise of the target)

- Target: Covid Vaccine | Text: “كل اللي اخذو تطعيم كورونا دخلو المرحله الثانيه في تحول الجينات الله يستر المرحله الثالثه يتحولون زومبي” → against (black-humor conspiracy mockery of the vaccine — against)
- Target: Covid Vaccine | Text: “السلام عليكم متى عمل مركز تطعيم كورونا في الظهران ؟” → none (pure inquiry, no hint of the author’s position)
- Target: Digital Transformation | Text: “#عاجل التعليم العالي | لا عودة عن التعليم الوجاهي للجامعات ولا صحة لما يتم تداوله عن التحول للتعليم الإلكتروني. #أحمد_” → none (news report relaying an announcement, no author position revealed)

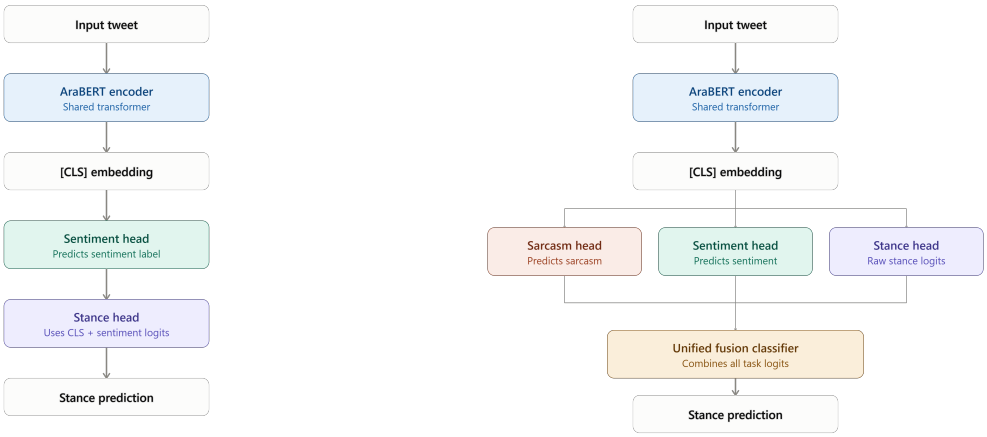
Input Data

- **Target:** `{topic}`
- **Text:** `{text}`

Output Format
Return ONLY a JSON object with the following keys:

- `"label"`: (The chosen stance: favor, against, none)
- `"reasoning"`: (A brief, 1-sentence explanation in English of why this label was chosen)
- `"confidence"`: (A score between 0 and 1)

E Multi-task Learning Model Architectures

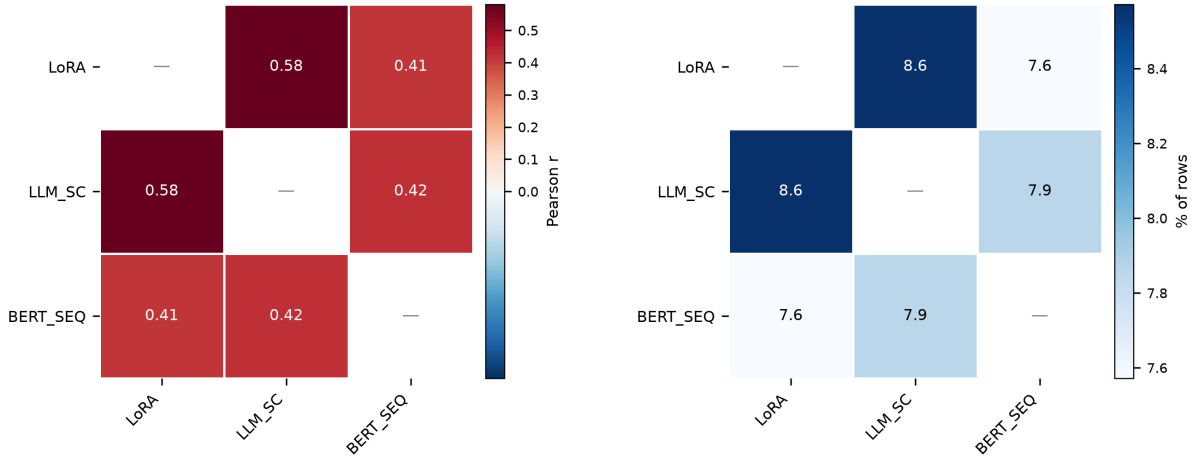


(a) Sequential multi-task learning model.

(b) Parallel multi-task learning model.

Figure 1: Architectures of our two best multi-task learning models.

F Ensemble Models Error Correlation Analysis



(a) Pairwise correlation of per-row correctness across the three ensembled models.

(b) Percentage of rows on which both models in each pair produced an incorrect prediction.

Figure 2: Error correlation across the three ensembled models: LoRA (fine-tuned Qwen2.5-14B-Instruct), LLM_SC (self-consistency with escalation), and BERT_SEQ (sequential multi-task AraBERTv0.2-Twitter-base).

G Error Analysis

Category	n	Gemini Err.	Ensemble Err.
Matched	1,051	6.1%	6.0%
Mismatched	349	33.2%	30.1%
All rows	1,400	12.9%	12.0%

Table 4: Development-set error rate by sentiment–stance match type, for our two best submissions. *None*-stance rows are paired with *Neutral* sentiment as their own matched case.

H Encoder-based Methods Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Batch size	32
Max sequence length	128
Learning rate	2e-5
Epochs (max)	20
Early stopping patience	3
Loss function	Cross-Entropy

Table 5: Hyperparameters used for baseline model training (Section 4.1).

Parameter	Description	Sequential	Parallel
Auxiliary task(s)	Auxiliary task(s) integrated with stance	sentiment	sarcasm + sentiment
Task order / fusion	How auxiliary signal reaches the stance head	sentiment → stance (logit concat.)	parallel heads → unified fusion
Head architecture	Architecture of each task head	2-layer MLP (Linear–Tanh–Dropout–Linear)	2-layer MLP (Linear–Tanh–Dropout–Linear)
Detach auxiliary logits	Whether auxiliary logits are detached before downstream use	False	N/A
Task weights	Relative weight of each task's loss	sentiment = 1.0, stance = 1.0	sarcasm = 0.1, sentiment = 0.2, stance = 0.7
Loss function	Loss applied per task	Per-task contrastive loss	Per-task contrastive loss (class-balanced, $\beta = 0.99$ for sarcasm & stance)
Margin	Contrastive loss margin	0.5	0.5
$\lambda_{\text{contrastive}}$	Weight of the contrastive term	1.0	1.0
CLS in fusion	Whether the fusion layer also sees the CLS embedding	N/A	False
Confidence gating	Whether auxiliary logits are scaled by prediction confidence	N/A	False
Batch size	Training batch size	32	32
Learning rate	Optimizer learning rate	2e-5	2e-5
Epochs (max)	Maximum training epochs	12	20
Early stopping patience	Epochs without improvement before stopping	5	5
Max sequence length	Token truncation/padding length	128	128

Table 6: Hyperparameters for the best sequential and parallel multi-task configurations.

I LoRA Fine-tuning Hyperparameters

Hyperparameter	Value
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	2
Learning rate	2×10^{-4}
Batch size	2
Gradient accumulation	8
Effective batch size	16

Table 7: LoRA fine-tuning hyperparameters.